

浙江大学

Zhejiang University-University of Illinois Urbana-Champaign Institute

Undergraduate Thesis (Design)



Thesis Title	Adaptive Federated Learning Over the Air
Student Name	Peiran Wei
Student ID	3220111228
Supervisor	Hao Yang
Class of year	2026
Major	ECE
Submission date	2026.05.22

Content Checklist

(Student should bind this form on the first page of the thesis after the cover page. The form should be filled out by the supervisor after the oral defense.)

Student Name	Peiran Wei	ZJU ID	3220111228	Major	ECE
Supervisor	Hao Yang			Academic Title	Assistant Professor
Thesis title	Adaptive Federated Learning Over the Air				
Content Checklist					Evaluation Section (√)
Senior Thesis	1. Cover (Use the unified cover, blue recommended)			Good	Fair
	2. Academic Integrity Commitment of Senior Thesis				
	3. Acknowledgements				
	4. Abstract				
	5. Table of Contents				
	6. Main Content (Page numbers start from the main text)				
	7. Reference				
	8. Appendix (if need)				
	9. Author's Biography				
	10. Task plan of senior thesis				
	11. Assessment Form of senior thesis				
	12. On-site Minutes Form of Oral Defense				
Signature of Supervisor:				Date:	

**ZJU-UIUC INSTITUTE**Zhejiang University-University of Illinois Urbana-Champaign Institute
浙江大学伊利诺伊大学厄巴纳香槟校区联合学院

Academic Integrity Commitment of Senior Thesis

1. I solemnly promise that the submitted thesis has been completed under the guidance of my supervisor in strict accordance with the relevant regulations of the university and the college.
2. In my thesis, except for the parts specifically marked and acknowledged, it does not contain research results that have been published or written by others, nor does it contain materials used to obtain degrees or certificates from Zhejiang University or other educational institutions.
3. Any contributions made by my colleagues to this research have been clearly stated and acknowledged in the thesis.
4. I promise that there has been no fabrication of data in the process of completing the thesis.
5. If there is any infringement of intellectual property rights in this thesis, I shall bear the corresponding legal responsibilities.
6. I fully understand that Zhejiang University has the right to retain and submit copies and disks of this thesis (design) to relevant departments or agencies, and to allow the thesis (design) to be consulted and borrowed. I authorize Zhejiang University to include all or part of this thesis (design) in relevant databases for retrieval and dissemination, and to save and compile this thesis (design) by means of photocopying, reduction, or scanning.

Author's Signature:

Supervisor's Signature:

Date:

Date:

Acknowledgements

I would like to express my sincere gratitude to my supervisor Hao Yang and my important advisor Chenhao Wang for their guidance, feedback, and support throughout this project. Their advice helped shape the technical direction of the work and kept the thesis focused on a clear and achievable research goal.

I am also grateful to the ZJU-UIUC Institute for providing the academic environment and course support that made this project possible. Finally, I would like to extend my heartfelt gratitude to my family and friends for their support during the experimentation and writing stages of this thesis.

Abstract

Over-the-air federated learning (OTA-FL) scales with the number of clients by letting analog client updates superpose directly in the wireless multiple-access channel, but the same superposition injects channel noise into the server-side aggregate. When the interference is heavy-tailed, rare but very large impulsive samples can dominate a FedAvg-style update and destabilize training. This thesis studies whether server-side adaptive optimization can improve robustness by responding to noisy OTA aggregates: a fractional second-moment accumulator increases normalization on impulsive coordinates and thereby reduces their effective update scale without requiring channel state information (CSI).

We implement a PyTorch OTA-FL simulator with a symmetric α -stable channel oracle based on the Chambers–Mallows–Stuck sampler, three server optimizers (FedAvg-OTA, AdaGrad-OTA, Adam-OTA) that all consume the same noisy aggregate, Median Anchored Clipping (MAC) as a robust pre-processing option, and an experiment layer that produces JSON logs, figures, and LaTeX tables. The parameter α is the tail index of the wireless noise: smaller α indicates a heavier tail and poorer channel conditions, while the AWGN case is recovered at $\alpha = 2$.

On CIFAR-10 (ResNet-18, $N = 100$, 100 rounds, $\gamma = 0.05$, 3 seeds), AdaGrad-OTA completes all runs and its mean final accuracy increases from 22.12% to 35.35% as α rises from 1.1 to 1.9, while Adam-OTA also completes all runs and improves from 28.64% to 39.65%. This trend matches the channel interpretation: larger α means a lighter-tailed, less impulsive channel. In contrast, FedAvg-OTA completes no runs for $\alpha \leq 1.9$. At the AWGN endpoint $\alpha = 2$, all methods complete, with Adam-OTA reaching 39.74% mean final accuracy. MAC at the default operating point ($\alpha = 1.3$, $\gamma = 0.05$, $k = 3$) is empirically neutral within seed variation. A targeted server learning-rate sweep under the severe $\alpha = 1.3$ channel shows that Adam-OTA is tuning-sensitive: reducing

η from 10^{-2} to 10^{-3} improves its mean final accuracy from 26.70% to 33.65%, while AdaGrad-OTA reaches 34.62% at $\eta = 10^{-2}$. The project is therefore not a complete reproduction of the reference paper; its contribution is a reproducible empirical validation of a carefully scoped adaptive OTA-FL robustness claim, together with a finite-trace stability analysis of the noise-adaptive normalization mechanism.

Keywords: federated learning; over-the-air aggregation; α -stable noise; heavy-tailed interference; adaptive optimization; Median Anchored Clipping

Content

Chapter 1. Introduction	2
1.1 Motivation	2
1.2 Problem Statement	2
1.3 Contributions	3
1.4 Thesis Organization	4
Chapter 2. Background and Related Work	5
2.1 Federated Learning	5
2.2 Analog Over-the-Air Aggregation	5
2.3 Adaptive Gradient Methods	6
2.4 Robust Aggregation under Impulsive Noise	6
2.5 Positioning Relative to the Reference Paper	6
Chapter 3. System Design and Methodology	8
3.1 Training Loop	8
3.2 Median Anchored Clipping	8
3.3 Server-Side Baselines	9
3.4 AdaGrad-OTA	10
3.5 Adam-OTA	10
3.6 Finite-Trace Stability Analysis	11
3.7 Implementation Modules	14
Chapter 4. Experiments and Results	15
4.1 Data	15
4.2 Experiment Configuration	15
4.3 Evaluation Metrics	16
4.4 Main Method Comparison	17
4.5 Alpha-Stable Tail-Index Ablation	19
4.6 Server Learning-Rate Sweep	20
4.7 MAC Robust Pre-Processing	22
4.8 Client-Count Ablation	24
4.9 Discussion	25
4.9.1 Findings	25
4.9.2 Failures and Negative Results	26
4.9.3 Threats to Validity	26

Chapter 5. Conclusion and Future Work	27
5.1 Conclusion	27
5.2 Limitations	27
5.3 Future Work	28
References	29
Appendix	31
Author's Biography	33

Chapter 1. Introduction

1.1 Motivation

Federated learning (FL) trains a shared model across distributed clients without requiring those clients to upload raw data, which is attractive for edge intelligence because it reduces privacy exposure and keeps data near its source device [1]. The communication cost of conventional digital FL grows with the number of participating clients, because every client must send an update in a separate resource block. Analog over-the-air aggregation (OTA-FL) changes this by letting clients transmit updates as analog waveforms at the same time, so that the wireless multiple-access channel physically superposes the signals and the server observes a single aggregate in one shared channel use.

OTA-FL therefore scales in clients, but the same superposition also injects noise, interference, and possibly fading into the aggregate update. When the interference is heavy-tailed, a rare impulsive sample in a single round can be orders of magnitude larger than a typical client pseudo-gradient, which makes plain FedAvg-style updates oscillate or diverge. The reference work “Adaptive Federated Learning Over the Air” [2] studies this regime under fading and symmetric α -stable interference. This project follows that direction and makes the α -stable tail index a core experimental variable; AWGN is treated as the special case $\alpha = 2$.

1.2 Problem Statement

This thesis studies FL over a simulated analog OTA aggregation channel. At communication round t , the server receives a noisy aggregate update $\tilde{g}^t$:

$$\tilde{g}^t = \frac{1}{N\eta_{\text{loc}}} \left(\sum_{n=1}^N \Delta_n^t + \xi^t \right), \quad (1-1)$$

where N is the number of clients, Δ_n^t is the pseudo-gradient produced by client n , η_{loc} is the local SGD learning rate, and ξ^t is symmetric α -stable channel noise. Smaller α

produces heavier tails and more frequent impulsive outliers; $\alpha = 2$ recovers the Gaussian AWGN case.

The central research question is concrete and experimentally testable: how does the final test accuracy of FedAvg-OTA, AdaGrad-OTA, and Adam-OTA depend on the tail index α and the noise scale γ , when all three optimizers consume the same OTA aggregate $\tilde{g}^t$? The current codebase implements a controlled simulator for the core heavy-tail mechanism (α -stable OTA aggregation, adaptive server updates, and MAC robust pre-processing) and evaluates a narrower robustness claim.

1.3 Contributions

The contributions of this thesis are split into empirical findings and reusable artifacts.

Findings

- Under the implemented heavy-tail CIFAR-10 setting (ResNet-18, $N = 100$, 100 rounds, $\gamma = 0.05$, three seeds), AdaGrad-OTA and Adam-OTA complete all three seeds for every tested heavy-tail value, while FedAvg-OTA completes no runs for $\alpha \leq 1.9$. This directly supports the hypothesis that fractional second-moment adaptation stabilizes training in the presence of α -stable impulsive noise.
- The observed α -trend for the adaptive methods is broadly monotone: final accuracy improves as α approaches the Gaussian endpoint. This agrees with the reference paper’s qualitative claim that heavier tails slow training, although the absolute ranking between AdaGrad-OTA and Adam-OTA remains hyperparameter-dependent.
- A targeted learning-rate sweep at $\alpha = 1.3$ shows that Adam-OTA’s lower default result is partly an lr-mismatch effect: reducing η from 10^{-2} to 10^{-3} raises its mean final accuracy from 26.70% to 33.65%. AdaGrad-OTA is best at the default $\eta = 10^{-2}$ with 34.62%, so tuning narrows but does not reverse the ranking in the tested grid.

- At $\alpha = 1.3$, $\gamma = 0.05$, $k = 3$, Median Anchored Clipping is empirically neutral within seed variation. The MAC channel therefore demands a stronger operating point (smaller α , larger γ , or smaller k) before it can deliver a measurable accuracy gain.

Artifacts

- A modular PyTorch OTA-FL simulator with a synchronous FL training loop, four server optimizers that consume the same OTA aggregate, a symmetric α -stable channel oracle based on the Chambers–Mallows–Stuck sampler, MAC pre-processing, JSON result export, and plotting scripts.
- A finite-trace stability analysis that explains why a fractional second-moment accumulator contracts the effective coordinate-wise learning rate on impulsive rounds, together with a tail-index scaling argument for long-run step-size decay.

1.4 Thesis Organization

Chapter 2 reviews FL, OTA aggregation, and adaptive optimization, and positions this project against the reference paper. Chapter 3 describes the implemented system model and optimizer equations. Chapter 4 presents the datasets, experimental configuration, generated figures, and numerical results. Chapter 5 summarizes the outcome, limitations, and future work.

Chapter 2. Background and Related Work

2.1 Federated Learning

Federated learning considers the optimization of a global model over data distributed across multiple clients. If client n contributes local data $\mathcal{D}_n$, the global objective is

$$\min_w F(w) = \sum_{n=1}^N p_n F_n(w), \quad F_n(w) = \frac{1}{|\mathcal{D}_n|} \sum_{(x,y) \in \mathcal{D}_n} \ell(w; x, y), \quad (2-1)$$

where p_n is often proportional to the local data size $|\mathcal{D}_n|$. FedAvg solves this objective by alternating local client optimization with server-side averaging [1], and is communication-efficient compared with fully synchronous distributed SGD. Later work addresses client drift and data heterogeneity: FedProx adds a proximal term on the local objective [3]; SCAFFOLD uses control variates to correct the drift [4]; and Adaptive Federated Optimization shows that server-side adaptive methods such as FedAdaGrad and FedAdam can outperform plain FedAvg under the same communication budget [5]. A comprehensive treatment of open problems in FL is given by [6] and by [7].

2.2 Analog Over-the-Air Aggregation

Over-the-air computation exploits the physical superposition of simultaneously transmitted analog signals over the wireless multiple-access channel, so that the receiver directly observes a function (here, a sum) of client inputs rather than decoding each transmission separately [8, 9]. Applying this idea to FL yields OTA-FL: clients transmit analog updates concurrently, and the server aggregates in one shared channel use [10–14]. A simplified channel model, without fading or power control, is

$$y^t = \sum_{n=1}^N \Delta_n^t + \xi^t, \quad (2-2)$$

and the server uses $\tilde{g}^t = y^t/N$ as the update direction. In the AWGN case $\xi^t \sim \mathcal{N}(0, \sigma^2 I)$. In the heavier-tailed setting studied in this project, ξ^t follows a symmetric

α -stable distribution [15, 16].

2.3 Adaptive Gradient Methods

Adaptive optimizers adapt the learning rate separately for each variable based on historical gradients. AdaGrad accumulates squared gradients and scales each coordinate’s learning rate inversely to the square root of its accumulated magnitude [17]. Adam combines first-moment smoothing, second-moment exponential averaging, and bias correction [18]. Convergence of AdaGrad under adversarial noise was analyzed in [19], and a simple convergence proof of Adam under relaxed assumptions is given by [20]. In centralized learning these mechanisms are typically motivated by sparse gradients, noisy mini-batches, and heterogeneous coordinate scales.

In OTA-FL, the same mechanisms have an additional interpretation. If channel noise makes the aggregate update unusually large in some coordinate, the second-moment state for that coordinate increases. The denominator of the adaptive update then becomes larger, reducing the effective step size. This creates an implicit feedback loop from observed noisy update magnitude to the server learning rate. The project tests whether this feedback loop is visible in training curves and ablation studies.

2.4 Robust Aggregation under Impulsive Noise

Heavy-tailed and Byzantine perturbations motivate a parallel line of work on robust aggregation. Krum selects one gradient closest to its neighbours in ℓ_2 distance [21]; coordinate-wise median and trimmed mean tolerate up to a constant fraction of adversarial clients [22]; and robust aggregation by weighted geometric median gives communication-efficient Byzantine-robust FL [23]. Classical robust statistics gives the older foundations: Huber-type M-estimators and the median absolute deviation [24, 25] are exactly the primitives used in Median Anchored Clipping.

2.5 Positioning Relative to the Reference Paper

The reference paper proposes adaptive federated learning over the air and studies AdaGrad-like and Adam-like server updates with theoretical analysis under α -stable in-

terference [2]. It evaluates multiple image and character classification tasks and includes studies over tail index, client count, and data heterogeneity. The present project uses that paper as the conceptual source but narrows the deliverable. The implemented framework supports the core OTA-FL training loop, the Chambers–Mallows–Stuck α -stable sampler [26], MAC robust pre-processing, and four adaptive server optimizers. The main evidence is the α -stable tail-index ablation, with AWGN reported as the $\alpha = 2$ endpoint rather than the whole scope. The explicit scope difference is summarized in Table 2.1.

Table 2.1 Scope difference between this project and the reference paper [2].

Aspect	Reference paper	This project
Channel model	α -stable + Rayleigh fading	Symmetric α -stable only,
Tail index coupling	$h_{n,t}$ + power control	no fading, no power control
Datasets	Channel α and optimizer denominator α are independent	Channel α and denominator α are shared
Convergence analysis	CIFAR-10, CIFAR-100, FEMNIST and more	MNIST, CIFAR-10
Empirical scale	Full non-convex proof under α -stable noise	Finite-trace stability propositions plus toy intuition
Hardware validation	Per-method learning-rate tuning across many seeds and configurations	Three seeds, fixed server learning rate per experiment
Primary output	SDR testbed claimed in the paper	Pure software simulator
	Algorithmic contribution and convergence theorems	Reproducible simulator, finite-trace theory, and empirical robustness evidence

Chapter 3. System Design and Methodology

3.1 Training Loop

The simulator follows the standard synchronous FL round structure. At round t , the server broadcasts the global model w^t to all clients. Each client copies the global model, trains locally for E epochs using SGD with local learning rate η_{loc} , and returns a pseudo-gradient

$$\Delta_n^t = w^t - [w_n^t]^{(E)}, \quad (3-1)$$

where $[w_n^t]^{(E)}$ is the local weight after E epochs. Because the local update reduces F_n , we have $\Delta_n^t \approx \eta_{\text{loc}} \sum_{e=1}^E \nabla F_n(w_{n,e-1}^t)$; that is, Δ_n^t points in the same direction as the accumulated local gradient, so the server update $w_{t+1} = w_t - \eta \tilde{g}^t$ moves along a descent direction of F up to channel noise. The OTA oracle aggregates these pseudo-gradients:

$$\tilde{g}^t = \frac{1}{N\eta_{\text{loc}}} \left(\sum_{n=1}^N \Delta_n^t + \xi^t \right), \quad \xi^t \sim S\alpha S(\gamma), \quad (3-2)$$

where $S\alpha S(\gamma)$ denotes a symmetric α -stable random vector with tail index α and scale γ . The division by η_{loc} converts the local parameter difference back into an average-gradient scale, matching the implementation in `src/experiments/federated.py`. The special case $\alpha = 2$ corresponds to AWGN with standard deviation $\sqrt{2}\gamma$ under the sampler convention. The server optimizer receives $\tilde{g}^t$ and updates the global model in place.

3.2 Median Anchored Clipping

The simulator includes Median Anchored Clipping (MAC) as a robust pre-processing step before the server optimizer. For an aggregated vector $\tilde{g}^t \in \mathbb{R}^d$, MAC anchors the update at the coordinate-wise median m , sets the clipping radius to k times the Median

Absolute Deviation (MAD), and projects every coordinate into the resulting band:

$$m = \text{median}\{\tilde{g}_i^t : i \in \{1, 2, \dots, d\}\}, \quad (3-3)$$

$$\tau = k \cdot \text{median}\{|\tilde{g}_i^t - m| : i \in \{1, 2, \dots, d\}\}, \quad (3-4)$$

$$\tilde{g}_i^t \leftarrow m + \text{clip}(\tilde{g}_i^t - m, -\tau, +\tau). \quad (3-5)$$

The first equation defines the anchor, the second sets the clipping radius τ as k times the MAD of the aggregate, and the third clips each coordinate's deviation from the anchor to the band $[-\tau, \tau]$ and adds the anchor back. k here serves as the clipping factor with default setting $k = 3$, which matches the implementation in `src/channel/mac.py`. MAC is expected to have little effect in the Gaussian endpoint, but it can reduce rare impulsive coordinates when $\alpha < 2$.

3.3 Server-Side Baselines

The plain OTA baseline applies the noisy aggregate directly:

$$w_{t+1} = w_t - \eta \tilde{g}^t. \quad (3-6)$$

The momentum baseline maintains a first-moment buffer:

$$m_t = \rho m_{t-1} + \tilde{g}^t, \quad w_{t+1} = w_t - \eta m_t. \quad (3-7)$$

These baselines are useful because they isolate whether the second-moment normalization in adaptive methods provides additional robustness beyond simple smoothing.

3.4 AdaGrad-OTA

AdaGrad-OTA maintains a smoothed update direction s_t and an accumulated coordinate-wise magnitude v_t :

$$s_t = \beta_1 s_{t-1} + (1 - \beta_1) \tilde{g}^t, \quad (3-8)$$

$$v_t = v_{t-1} + |s_t|^\alpha, \quad (3-9)$$

$$w_{t+1} = w_t - \eta \frac{s_t}{v_t^{1/\alpha} + \varepsilon}. \quad (3-10)$$

For $\alpha = 2$, the denominator reduces to the usual square-root AdaGrad normalization. For $\alpha < 2$, the same form follows the fractional α -stable accumulation rule used in the reference paper. Because v_t is monotone increasing, AdaGrad-OTA tends to become conservative over long runs. This can stabilize noisy training, but it may also slow late-stage convergence.

3.5 Adam-OTA

The implemented Adam-OTA variant uses a stable server-side Adam update on the OTA-aggregated direction:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \tilde{g}^t, \quad (3-11)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) |\tilde{g}^t|^\alpha, \quad (3-12)$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad (3-13)$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}, \quad (3-14)$$

$$w_{t+1} = w_t - \eta \frac{\hat{m}_t}{\hat{v}_t^{1/\alpha} + \varepsilon}. \quad (3-15)$$

When $\alpha = 2$, this is classical Adam applied at the server to the noisy OTA aggregate. For $\alpha < 2$, the second-moment update becomes a fractional moment accumulator. This preserves the project hypothesis because the update uses a state variable to compress effective step sizes when the observed aggregate has large magnitude.

Implementation note: ε . In the code, AdaGrad-OTA uses $\varepsilon = 10^{-4}$ while Adam-OTA uses $\varepsilon = 10^{-8}$. The two differ by four orders of magnitude by design: AdaGrad-OTA has no β_2 exponential moving average, so its first-round v_t is very small and a larger ε is needed to damp the early transient; Adam-OTA's bias correction $\hat{v}_t = v_t/(1 - \beta_2^t)$ delivers a non-negligible denominator already after a few rounds, so the smaller textbook default $\varepsilon = 10^{-8}$ is kept. The asymmetry between the two optimizers is the dominant reason why their early-round test-loss transients in Figure 4.1 differ in shape.

3.6 Finite-Trace Stability Analysis

The project does not reproduce the full non-convex convergence proof in [2]. It does, however, admit a rigorous finite-trace analysis that matches the implemented simulator. The analysis is coordinate-wise and concerns the denominator that turns a noisy OTA aggregate into an adaptive server step. For a fixed coordinate i , let z_t denote the quantity accumulated by the optimizer: $z_t = s_{t,i}$ for AdaGrad-OTA and $z_t = \tilde{g}_i^t$ for Adam-OTA. Define $x_t = |z_t|$.

Moment caveat. For a non-Gaussian symmetric α -stable random variable ($0 < \alpha < 2$), $\mathbb{E}|\xi|^p$ is finite only for $0 < p < \alpha$ and is infinite for $p \geq \alpha$ [15]; the endpoint $\alpha = 2$ is the Gaussian case. Therefore, in the heavy-tailed regime $\alpha < 2$, the usual bounded-variance SGD argument is not the right tool, and even the population α -moment used by the fractional accumulator need not exist. The implemented algorithms instead operate on a finite realized training trace, where every stored accumulator is finite unless numerical overflow occurs. The propositions below should be read in that finite-trace sense.

Proposition 1: impulse contraction. For AdaGrad-OTA, define

$$V_t = \sum_{\tau=1}^t x_\tau^\alpha, \quad \eta_{t,i}^{\text{eff}} = \frac{\eta}{V_t^{1/\alpha} + \varepsilon}. \quad (3-16)$$

If round t contains an impulsive coordinate with magnitude $x_t = R$, then

$$\eta_{t,i}^{\text{eff}} \leq \frac{\eta}{R + \varepsilon}, \quad \left| \eta \frac{z_t}{V_t^{1/\alpha} + \varepsilon} \right| \leq \eta. \quad (3-17)$$

Thus the same observation that creates a large numerator also enlarges the denominator enough to cap the coordinate-wise normalized step. A fixed-step FedAvg-OTA update has no such feedback: its coordinate response to the same impulse has magnitude ηR .

Proof. Since $V_t = V_{t-1} + R^\alpha \geq R^\alpha$, we have $V_t^{1/\alpha} \geq R$. Substituting this inequality into the effective learning rate gives the first bound, and multiplying by R gives the normalized-step bound.

For Adam-OTA, the bias-corrected second accumulator can be written as

$$\hat{V}_t = \sum_{\tau=1}^t \frac{(1 - \beta_2)\beta_2^{t-\tau}}{1 - \beta_2^t} x_\tau^\alpha. \quad (3-18)$$

The current sample has weight $(1 - \beta_2)/(1 - \beta_2^t) \geq 1 - \beta_2$. Therefore, after an impulse $x_t = R$,

$$\frac{\eta}{\hat{V}_t^{1/\alpha} + \varepsilon} \leq \frac{\eta}{(1 - \beta_2)^{1/\alpha} R + \varepsilon}. \quad (3-19)$$

Adam-OTA keeps a finite memory rather than a monotone memory, so this bound contains the constant $(1 - \beta_2)^{1/\alpha}$, but the same denominator feedback remains present.

Proposition 2: tail-index step-size scaling. For AdaGrad-OTA, suppose that along a finite trace the empirical α -power average is bounded below by a positive constant,

$$c_t = \frac{1}{t} \sum_{\tau=1}^t x_\tau^\alpha \geq c > 0, \quad 1 \leq t \leq T. \quad (3-20)$$

Then each coordinate-wise effective learning rate satisfies

$$\eta_{t,i}^{\text{eff}} \leq \eta c^{-1/\alpha} t^{-1/\alpha}, \quad (3-21)$$

and its average over the first T rounds obeys

$$\frac{1}{T} \sum_{t=1}^T \eta_{t,i}^{\text{eff}} \leq \eta c^{-1/\alpha} \left(\frac{1}{T} + \frac{\alpha}{\alpha-1} T^{-1/\alpha} \right). \quad (3-22)$$

Proof. The denominator satisfies $V_t^{1/\alpha} = (tc_t)^{1/\alpha} \geq (tc)^{1/\alpha}$, so $\eta_{t,i}^{\text{eff}} \leq \eta c^{-1/\alpha} t^{-1/\alpha}$. Averaging this bound and using $\sum_{t=1}^T t^{-1/\alpha} \leq 1 + \int_1^T u^{-1/\alpha} du \leq 1 + \frac{\alpha}{\alpha-1} T^{1-1/\alpha}$ gives the result.

This proposition gives a direct tail-index interpretation. Smaller α produces the faster decay exponent $1/\alpha$, so heavy-tailed traces force stronger automatic damping. That damping improves stability after impulsive rounds, but it can also compress useful signal and slow late-stage optimization.

One-dimensional interpretation. The same mechanism can be visualized with the quadratic problem

$$f(w) = \frac{1}{2}(w - w^*)^2. \quad (3-23)$$

The noiseless gradient is $w_t - w^*$, while the OTA server observes

$$\tilde{g}_t = w_t - w^* + \xi_t, \quad \xi_t \sim S\alpha S(\gamma). \quad (3-24)$$

For an AdaGrad-style update,

$$v_t = v_{t-1} + |\tilde{g}_t|^\alpha, \quad w_{t+1} = w_t - \eta \frac{\tilde{g}_t}{v_t^{1/\alpha} + \epsilon}. \quad (3-25)$$

Whenever an impulsive noise sample makes $|\tilde{g}_t|$ large, it also makes v_t larger. The next effective step size

$$\eta_t^{\text{eff}} = \frac{\eta}{v_t^{1/\alpha} + \epsilon} \quad (3-26)$$

therefore contracts automatically. This special case is not a replacement for the full theorem in [2], but it explains why the implemented accumulator reacts strongly after

impulsive rounds.

3.7 Implementation Modules

The codebase is organized around separable modules. The channel layer contains the noisy aggregation oracle and noise sampler. The data layer contains dataset loading and client partitioning. The model layer contains classification networks. The optimizer layer contains the four server optimizers. The experiment layer contains single-run training, method comparison, ablation, plotting, logging and result export.

This organization supports the project goal in two ways. First, the α -stable experiments and MAC comparisons can be run reproducibly through command-line scripts. Second, the same interfaces can later be used to add richer physical channel models, additional datasets, or stricter reproduction settings without rewriting the training loop.

Chapter 4. Experiments and Results

4.1 Data

The experiments use the two datasets supported by the current codebase: MNIST and CIFAR-10. Both datasets are stored under the project-local `data/` directory and are loaded through the PyTorch `torchvision` dataset interfaces with `download=True`. No physical wireless-channel dataset is required, because the channel perturbation is generated synthetically by the OTA aggregation oracle.

MNIST is used as a lightweight sanity benchmark. It contains 60,000 training images and 10,000 test images of handwritten digits. The raw files are stored under `data/MNIST/raw/`. Images are converted to tensors and normalized with mean 0.1307 and standard deviation 0.3081.

CIFAR-10 is used as the main image-classification benchmark. It contains 50,000 training images and 10,000 test images across 10 object classes. The raw batch files `data_batch_1` through `data_batch_5`, `test_batch`, and `batches.meta` are stored under `data/cifar-10-batches-py/`. During training, CIFAR-10 uses random cropping with padding 4 and random horizontal flipping, followed by tensor conversion and channel-wise normalization. During testing, only tensor conversion and normalization are applied.

Federated clients are generated by partitioning the training set. The main comparison experiments use a non-IID Dirichlet partition with concentration 0.1 and random seed 42. In the generated partition, the MNIST comparison uses 10 clients, with client sizes ranging from 937 to 11,604 samples. The CIFAR-10 comparison uses 100 clients, with client sizes ranging from 12 to 1,733 samples. The single MNIST sanity run uses an IID split over 10 clients, with 6,000 samples per client.

4.2 Experiment Configuration

Table 4.1 summarizes the controlled comparison configuration. In these experiments all four server optimizers share the same dataset, model, partition, random seed, OTA

noise setting, and training budget. The OTA channel uses $\alpha = 2$, which corresponds to AWGN in the implemented sampler, and $\gamma = 0.05$.

Table 4.1 Controlled comparison configuration

Item	MNIST comparison	CIFAR-10 comparison
Dataset	MNIST	CIFAR-10
Model	MLP	ResNet-18
Optimizers	FedAvg-OTA, AdaGrad-OTA, Adam-OTA	FedAvg-OTA, AdaGrad-OTA, Adam-OTA
Number of clients N	10	100
Communication rounds	100	200
Local epochs E	5	5
Batch size	64	64
Server learning rate η	0.01	0.01
Local learning rate	0.01	0.01
Momentum / β_1	0.9	0.9
Adam β_2	0.999	0.999
Adam-OTA ε	10^{-8}	10^{-8}
AdaGrad-OTA ε	10^{-4}	10^{-4}
Noise tail index α	2.0	2.0
Noise scale γ	0.05	0.05
Partition	non-IID Dirichlet	non-IID Dirichlet
Dirichlet concentration	0.1	0.1
Random seed	42	42

Table 4.2 summarizes the revised CIFAR-10 heavy-tail ablation configuration. The tail-index ablation varies α while fixing $\gamma = 0.05$. The MAC comparison uses the same heavy-tailed channel with $\alpha = 1.3$ and compares training with and without robust pre-processing. The learning-rate sweep holds the same $\alpha = 1.3$ channel fixed and varies the server learning rate for AdaGrad-OTA and Adam-OTA.

4.3 Evaluation Metrics

The main metrics are test loss and test accuracy after each communication round. For comparison experiments, the final-round accuracy and the best accuracy achieved during the run are both reported. The final value measures the state of the trained model at the end of the fixed communication budget, while the best value helps reveal short-lived convergence peaks.

Table 4.2 CIFAR-10 ablation configuration

Item	Value
Dataset / model	CIFAR-10 / ResNet-18
Optimizers in alpha study	FedAvg-OTA, AdaGrad-OTA, Adam-OTA
Alpha values	1.1, 1.3, 1.5, 1.7, 1.9, 2.0
MAC comparison	no MAC vs MAC, $k = 3$
Learning-rate sweep	AdaGrad-OTA / Adam-OTA at $\alpha = 1.3$, $\eta \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1\}$
Communication rounds	100
Local epochs E	1
Batch size	64
Server learning rate η	0.001
Local learning rate	0.01
Momentum / β_1	0.9
Adam β_2	0.999
Adam-OTA ε	10^{-8}
AdaGrad-OTA ε	10^{-4}
Noise scale γ	0.05
Partition	non-IID Dirichlet, concentration 0.1
Random seeds	42, 43, 44

For the alpha ablation, the reported metric is final test accuracy under each tail index α . For the learning-rate sweep, the reported metric is final test accuracy for each server learning rate. For the MAC comparison, the reported metric is final test accuracy with and without clipping. Multi-seed results are reported as mean $\pm$ standard deviation. All revised heavy-tail values are generated from the JSON files under `results/heavytail/` rather than copied from a single run.

4.4 Main Method Comparison

The MNIST comparison in Table 4.3 is used as a sanity benchmark for adaptive server-side normalization. FedAvg-OTA reaches 67.85% final accuracy, while AdaGrad-OTA and Adam-OTA reach 93.05% and 93.46%, respectively. This supports the hypothesis that second-moment adaptation can stabilize OTA training in this lighter benchmark, but it should not be treated as the main heavy-tail evidence.

The CIFAR-10 comparison in Table 4.4 is more nuanced. AdaGrad-OTA and Adam-OTA improve over plain FedAvg-OTA under this specific CIFAR-10 configuration. There-

Table 4.3 MNIST comparison results ($N=10$, AWGN $\alpha=2$, $\gamma=0.05$, seed 42, 100 rounds, $E=5$, $\eta=0.01$; single seed, no error bar).

Method	Final loss	Final accuracy	Best accuracy
FedAvg-OTA	1.5018	67.85%	67.85%
AdaGrad-OTA	0.2091	93.05%	93.19%
Adam-OTA	0.2227	93.46%	93.46%

fore, the CIFAR-10 result should not be stated as “Adam-OTA is always best”. A more accurate conclusion is that adaptive methods help relative to FedAvg-OTA in this setting. This AWGN endpoint motivates the heavier α -stable experiments below: Gaussian noise alone is not the strongest stress test, so the project’s distinctive question is whether adaptive normalization and MAC help when the channel has impulsive tails.

Table 4.4 CIFAR-10 comparison results ($N=100$, AWGN $\alpha=2$, $\gamma=0.05$, seed 42, 200 rounds, $E=5$, $\eta=0.01$; single seed, no error bar).

Method	Final loss	Final accuracy	Best accuracy
FedAvg-OTA	0.9670	65.38%	65.38%
AdaGrad-OTA	0.8706	69.27%	69.27%
Adam-OTA	0.9278	67.41%	68.08%

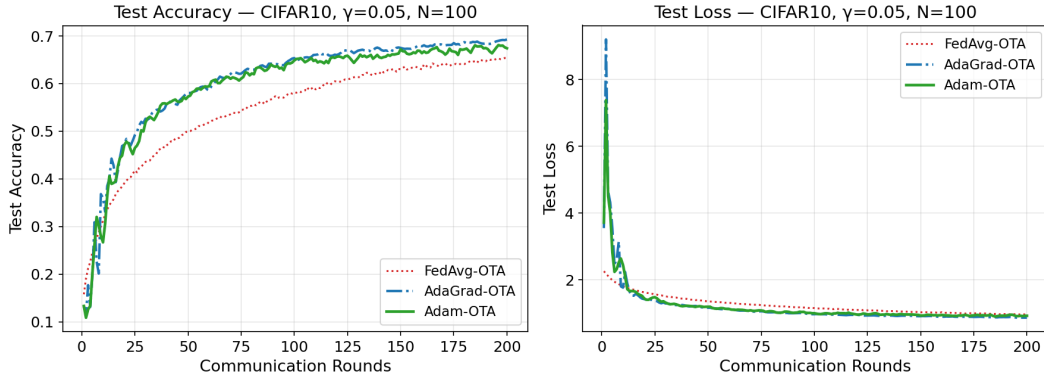


Figure 4.1 CIFAR-10 comparison curves for FedAvg-OTA, AdaGrad-OTA, and Adam-OTA under AWGN. The comparison focuses on the plain FedAvg baseline and the two adaptive server optimizers.

The single-run outputs provide an additional sanity check, but they are not a controlled four-method comparison because their hyperparameters differ from Table 4.1. The single CIFAR-10 Adam-OTA run uses batch size 128 and server learning rate 0.01, reaching 61.51% final accuracy. The single MNIST AdaGrad-OTA run uses an IID

client split and reaches 97.57% final accuracy. These runs confirm that the implemented optimizers can train successfully, but the main conclusions should be drawn from the controlled comparison tables above.

4.5 Alpha-Stable Tail-Index Ablation

The revised core experiment varies the α -stable tail index instead of only varying the AWGN scale. The endpoint $\alpha = 2$ is Gaussian AWGN, while smaller values of α produce heavier-tailed impulsive interference. This is the setting where adaptive normalization is most meaningful: a server method must handle rare but very large aggregate perturbations. The adaptive methods complete the full 100 communication rounds for all three seeds. In contrast, FedAvg-OTA often diverges under the heaviest-tailed settings; the generated table therefore reports the number of successful seeds rather than silently averaging failed runs.

Table 4.5 CIFAR-10 alpha-stable tail-index ablation: final accuracy, mean $\pm$ std over seeds

α	FedAvg-OTA	AdaGrad-OTA	Adam-OTA
1.1	– (0/3 ok)	22.12% $\pm$ 1.02%	28.64% $\pm$ 1.68%
1.3	– (0/3 ok)	25.29% $\pm$ 0.88%	33.87% $\pm$ 0.73%
1.5	– (0/3 ok)	28.74% $\pm$ 1.05%	36.41% $\pm$ 0.66%
1.7	– (0/3 ok)	31.96% $\pm$ 0.86%	38.25% $\pm$ 0.67%
1.9	– (0/3 ok)	35.35% $\pm$ 0.50%	39.65% $\pm$ 0.73%
2	21.43% $\pm$ 1.09%	36.67% $\pm$ 1.10%	39.74% $\pm$ 0.69%

Table 4.5 shows a clear stability separation between methods. FedAvg-OTA produces no successful runs for $\alpha \leq 1.9$ under the finite-value guard. At the Gaussian endpoint $\alpha = 2$, the baseline completes all seeds and reaches 21.43% mean final accuracy. In contrast, both adaptive methods complete all seeds for every tested α . AdaGrad-OTA improves from 22.12% at $\alpha = 1.1$ to 35.35% at $\alpha = 1.9$ and 36.67% at $\alpha = 2$. Adam-OTA improves from 28.64% at $\alpha = 1.1$ to 39.65% at $\alpha = 1.9$ and 39.74% at $\alpha = 2$.

The result supports the core robustness claim: server-side fractional second-moment normalization makes the implemented OTA-FL loop much less fragile under impulsive α -stable noise than FedAvg-style updates. It also matches the reference paper’s qualita-

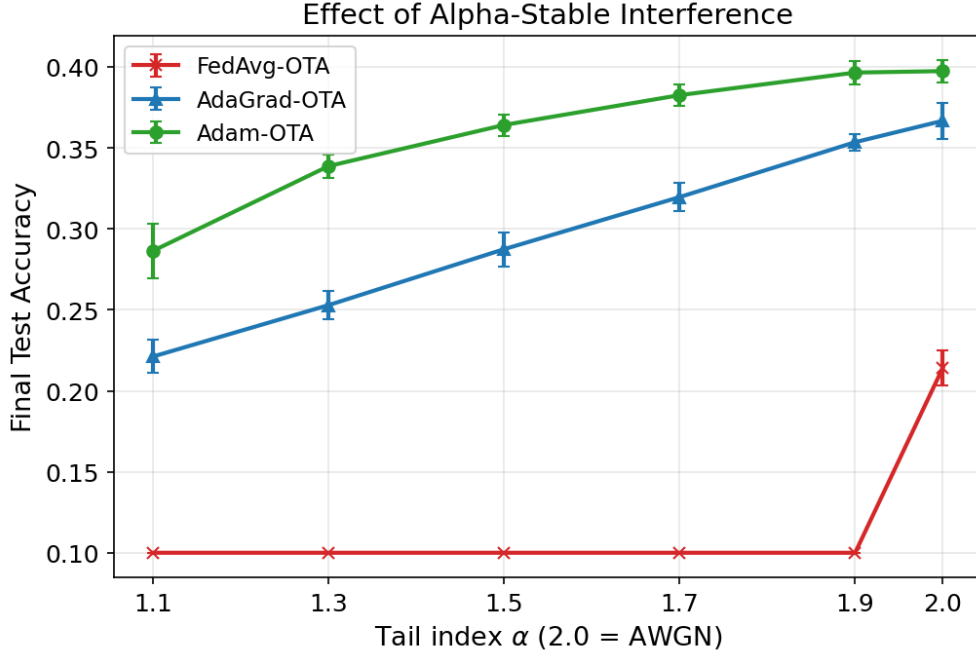


Figure 4.2 CIFAR-10 final accuracy as a function of α -stable tail index α .

tive tail-index message: heavier tails make training harder, and performance generally improves as α approaches the Gaussian endpoint. The result is still narrower than a full reproduction of [2]. The implementation omits fading and power control, uses a shared α for both the channel and the adaptive denominator, and uses a single global server learning rate rather than per-method tuning in this table. The next section runs a targeted learning-rate sweep to test how sensitive the $\alpha = 1.3$ comparison is to the shared learning-rate choice.

4.6 Server Learning-Rate Sweep

The tail-index ablation uses a single server learning rate, $\eta = 10^{-3}$, for all methods. To check sensitivity to this shared learning-rate choice at $\alpha = 1.3$, we rerun AdaGrad-OTA and Adam-OTA on the same CIFAR-10 / ResNet-18 setting over seven learning rates from 10^{-4} to 10^{-1} .

Table 4.6 and Figure 4.3 show that the adaptive methods prefer different server learning rates. AdaGrad-OTA is best at the default $\eta = 10^{-2}$, reaching $34.62\% \pm 0.64\%$ final accuracy, which matches the 34.56% result in the tail-index table up to rerun variation.

Table 4.6 Per-method server learning-rate sweep on CIFAR-10 at $\alpha = 1.3$, $\gamma = 0.05$: final accuracy, mean $\pm$ std over seeds

η	AdaGrad-OTA	Adam-OTA
0.0001	13.94% $\pm$ 1.02%	20.51% $\pm$ 1.40%
0.0003	18.77% $\pm$ 1.36%	26.64% $\pm$ 1.15%
0.001	25.33% $\pm$ 0.88%	33.65% $\pm$ 0.85%
0.003	32.28% $\pm$ 1.24%	32.69% $\pm$ 0.83%
0.01	34.62% $\pm$ 0.64%	26.70% $\pm$ 0.62%
0.03	29.82% $\pm$ 0.48%	21.20% $\pm$ 0.94%
0.1	– (0/3 ok)	– (0/3 ok)

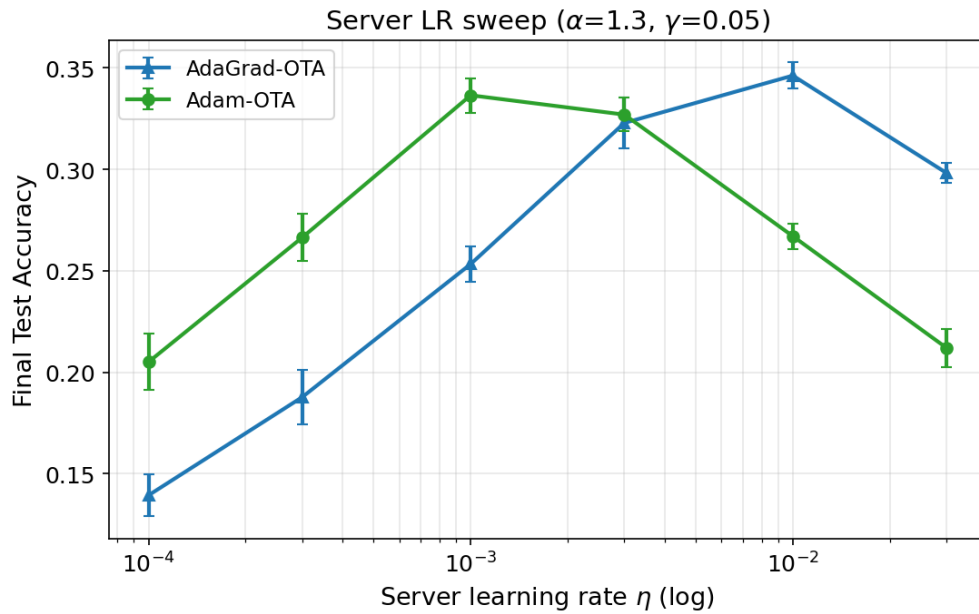


Figure 4.3 CIFAR-10 final accuracy as a function of server learning rate at $\alpha = 1.3$, $\gamma = 0.05$.

Adam-OTA is best one decade lower, at $\eta = 10^{-3}$, where it reaches $33.65\% \pm 0.85\%$. This is a 6.95 percentage-point gain over the $\eta = 10^{-2}$ row in the same sweep, and a 6.74 percentage-point gain over the default Adam-OTA result in Table 4.5. The sweep therefore shows that Adam-OTA’s lower default accuracy is partly a learning-rate mismatch, but the tested grid does not reverse the ranking: AdaGrad-OTA remains slightly ahead at its own best learning rate. Both methods fail the finite-value guard at $\eta = 10^{-1}$, confirming that the adaptive denominator does not make arbitrarily large server steps safe.

4.7 MAC Robust Pre-Processing

Median Anchored Clipping (MAC) is a robust pre-processing technique proposed by Li et al. [27] specifically for OTA-FL under heavy-tailed α -stable channel noise. It operates on the server-side aggregated gradient $\tilde{g}^t$ before the optimizer step: (1) compute the coordinate-wise median m , (2) center the gradient as $\tilde{g}^t - m$, (3) clip each centered coordinate to $[-k \cdot \text{MAD}, +k \cdot \text{MAD}]$ where $\text{MAD} = \text{median}(|\tilde{g}^t - m|)$, and (4) restore the median. Because the median is a robust statistic, MAC is far less sensitive to extreme outliers than gradient-norm clipping and preserves the proportional relationships among gradient dimensions [27].

From a theoretical standpoint, MAC is not expected to pair well with adaptive methods such as AdaGrad-OTA and Adam-OTA. MAC removes impulsive outliers before the optimizer sees them, whereas adaptive methods already provide implicit noise filtering: when a coordinate of $\tilde{g}^t$ is large, the fractional second-moment accumulator v_t grows, which automatically shrinks the effective step size for that coordinate in the next update. MAC clips the very signal that the adaptive denominator uses to gauge impulsiveness; consequently, the accumulator cannot contract the learning rate on rounds where the channel is noisy. The two mechanisms are therefore redundant at best and counter-productive at worst. For the non-adaptive baseline FedAvg-OTA, MAC may still help because this method lacks any built-in outlier shrinkage.

To test these predictions empirically, we compare training with and without MAC

under a heavy-tailed setting. The default comparison uses $\alpha = 1.3$, $\gamma = 0.05$, and clipping factor $k = 3$. MAC is not expected to change the Gaussian endpoint dramatically, but it should reduce the effect of rare impulsive coordinates when $\alpha < 2$.

Table 4.7 CIFAR-10 MAC comparison under heavy-tailed interference: final accuracy, mean $\pm$ std over seeds

Method	No MAC	MAC
FedAvg-OTA	11.91% $\pm$ 1.64%	11.87% $\pm$ 1.63%
AdaGrad-OTA	50.35% $\pm$ 0.40%	50.37% $\pm$ 0.66%
Adam-OTA	45.09% $\pm$ 0.74%	44.40% $\pm$ 1.27%

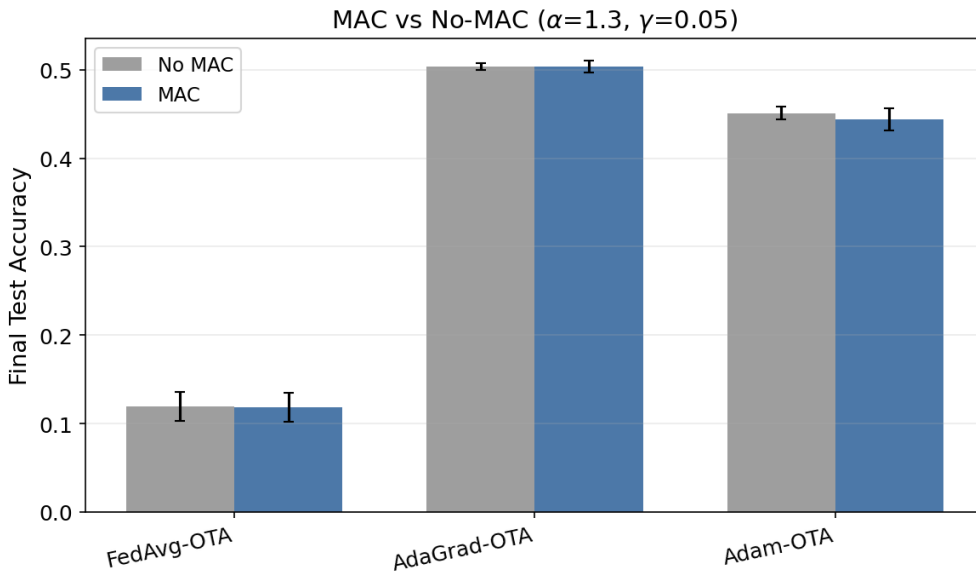


Figure 4.4 CIFAR-10 comparison of no-MAC and MAC under α -stable interference.

In the completed MAC comparison, clipping does not create a meaningful accuracy improvement. Across all three methods, the no-MAC and MAC results are within seed variation, so the current evidence is not strong enough to claim that MAC improves accuracy at this operating point. This is consistent with the theoretical discussion above: for adaptive methods, MAC clips the outliers before the fractional accumulator can measure them, so the adaptive denominator has less signal for contracting the step size. For non-adaptive baselines, MAC could in principle help, but at $\gamma = 0.05$ the empirical fraction of coordinates that violate the MAD band $[m - k \cdot \text{MAD}, m + k \cdot \text{MAD}]$ at $k = 3$ is small; MAC therefore has little signal to work on even for the baselines. A meaningful

MAC effect is expected to require a more impulsive operating point (smaller α , larger γ , or smaller k), which we explore in the `mac_sweep` extension in `src/experiments/heavy_tail.py`. MAC should therefore be reported as an implemented and tested robust pre-processing component, but the current evidence does not support a claim that MAC improves CIFAR-10 accuracy under the particular $\alpha = 1.3$, $\gamma = 0.05$, $k = 3$ operating point.

4.8 Client-Count Ablation

Table 4.8 reports the Adam-OTA client-count ablation on CIFAR-10. With the same 100-round budget, increasing the number of clients improves final accuracy from 10.26% at 5 clients to 21.72% at 100 clients. This trend is consistent with OTA aggregation: averaging more client updates can make the aggregate direction more informative, even though the data remain non-IID and the channel still injects noise.

This ablation uses the configuration in `src/experiments/ablation.py` (CIFAR-10, ResNet-18, AWGN $\alpha = 2$, $\gamma = 0.05$, 100 rounds, local epochs $E = 1$, batch size 64, server learning rate $\eta = 10^{-4}$, local learning rate 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.999$, non-IID Dirichlet 0.1, seed 42). The much smaller server learning rate, smaller local epochs, and shorter budget explain why the absolute Adam-OTA accuracies here are below Table 4.4, which uses $\eta = 0.01$, $E = 5$, and 200 rounds. The ablation isolates the effect of N rather than tuning a state-of-the-art Adam-OTA configuration.

Table 4.8 CIFAR-10 client-count ablation with Adam-OTA ($\eta = 10^{-4}$, $E = 1$, 100 rounds, $\alpha = 2$, $\gamma = 0.05$, $\beta_2 = 0.999$, non-IID Dirichlet 0.1, seed 42)

Clients N	Final loss	Final accuracy	Best accuracy
5	2.3121	10.26%	10.36%
10	2.2288	16.92%	16.96%
20	2.1249	19.99%	19.99%
50	2.1171	20.36%	20.36%
100	2.0880	21.72%	21.72%

Why does N not scale accuracy faster? In the implemented oracle, additive receiver noise is injected after the client updates are summed and then divided by $N\eta_{\text{loc}}$, so the

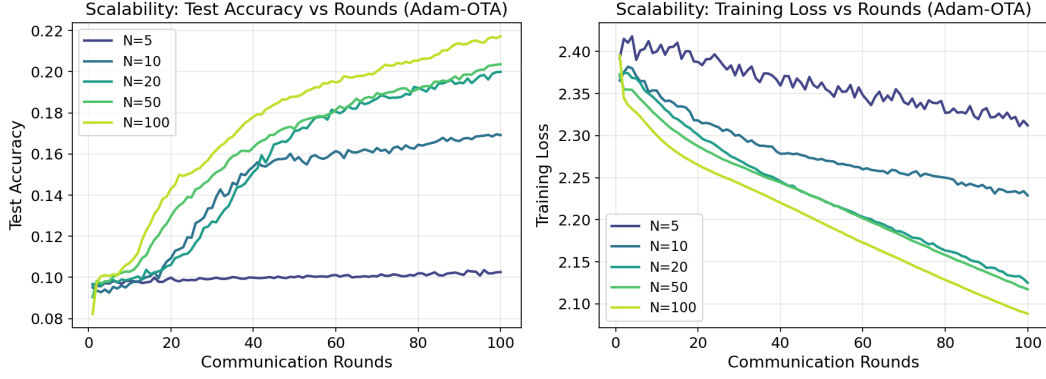


Figure 4.5 CIFAR-10 client-count learning curves for Adam-OTA.

channel-noise term decreases as $1/N$ in this simulator. The accuracy gain is nevertheless modest because increasing N also changes the non-IID client partition, reduces the average data per client, and keeps the same small server learning rate and 100-round budget. The client-count ablation should therefore be interpreted as a software sanity check that larger participation helps, not as a physical-layer scaling law.

4.9 Discussion

4.9.1 Findings

The experiments should be interpreted in two layers. The existing AWGN comparison remains a useful endpoint sanity check: it shows that the implementation trains successfully. The revised α -stable ablation is the central evidence for the project claim, because it tests the heavy-tailed channel condition that motivates adaptive OTA-FL in the first place.

The project conclusion should therefore be phrased carefully. The results do not prove that Adam-OTA universally dominates all baselines. Instead, the claim is that fractional second-moment adaptive updates provide a practical mechanism for improving OTA-FL stability when the aggregate update is corrupted by α -stable impulsive noise, and MAC provides a complementary but operating-point-dependent robustness lever.

4.9.2 Failures and Negative Results

Three observations are best reported as negative results rather than buried in caveats. (i) Adam-OTA does not universally dominate AdaGrad-OTA: although it is stronger in the current tail-index ablation, the targeted learning-rate sweep still finds an AdaGrad-OTA setting with slightly higher final accuracy at $\alpha = 1.3$. (ii) MAC at the default $\alpha = 1.3$, $\gamma = 0.05$, $k = 3$ operating point yields no measurable improvement within seed variation. (iii) The Adam-OTA CIFAR-10 result at $\alpha = 2$ in Table 4.4 (67.41%) is above FedAvg-OTA (65.38%) but below AdaGrad-OTA (69.27%), showing that second-moment normalization is not costless in the Gaussian regime.

4.9.3 Threats to Validity

The empirical evidence is bounded by four threats. First, seed counts differ across tables (three seeds for the α sweep and MAC comparison, single seed for the main comparison and client-count ablation); we note this in the captions rather than backfill all tables. Second, the learning-rate sweep is targeted rather than exhaustive: it covers only AdaGrad-OTA and Adam-OTA at $\alpha = 1.3$, so broader optimizer rankings across all α values, β_2 values, and baselines remain tuning-dependent. Third, the channel is a simulator, without fading, power control, or synchronization errors. Fourth, the strongest supported claim is empirical; the finite-trace stability analysis in §3.6 explains the implemented normalization mechanism but is not a full non-convex proof.

Chapter 5. Conclusion and Future Work

5.1 Conclusion

This thesis asks whether server-side adaptive optimization can act as an implicit, CSI-free filter for an α -stable OTA-FL channel. The main supported claim is empirical: under the three-seed CIFAR-10 tail-index ablation with $\gamma = 0.05$ and $\alpha \in \{1.1, 1.3, 1.5, 1.7, 1.9\}$, AdaGrad-OTA and Adam-OTA complete all runs, while FedAvg-OTA completes no runs for $\alpha \leq 1.9$. AdaGrad-OTA reaches 22.12–35.35% final accuracy across the heavy-tail sweep, and Adam-OTA reaches 28.64–39.65%. This supports the hypothesis that a fractional second-moment accumulator contracts the effective server learning rate on impulsive rounds and improves stability.

The claim is scoped rather than absolute. The observed α -trend for adaptive methods is broadly monotone and agrees with the qualitative message that heavier tails slow training. MAC at the default $\alpha = 1.3$, $\gamma = 0.05$, $k = 3$ operating point is empirically neutral; and the targeted $\alpha = 1.3$ learning-rate sweep shows that Adam-OTA improves from 26.70% to 33.65% when η is reduced from 10^{-2} to 10^{-3} , but still does not exceed AdaGrad-OTA’s best 34.62%, which shows that fractional second-moment normalization is not costless in the Gaussian regime. The main output is therefore a reproducible PyTorch simulator with α -stable tail-index experiments, MAC robustness comparisons, and a finite-trace stability analysis for the shrinkage mechanism, rather than a complete non-convex convergence proof or line-by-line reproduction of the reference paper.

5.2 Limitations

The project remains narrower than the full reference paper in four ways. First, it provides simulator-aligned finite-trace stability propositions rather than the full multi-dimensional non-convex convergence proof in the reference paper. Second, the wireless channel is a simulation oracle and does not model mobility, multi-cell interference, synchronization errors, CSI acquisition, or power control. Third, the project does not include a software-defined-radio or real wireless testbed validation. Fourth, the em-

pirical evidence is bounded by three seeds and a targeted learning-rate sweep at only one heavy-tailed operating point, so full optimizer rankings remain tuning-dependent (see §4.6 and §4.9.3).

5.3 Future Work

The most direct next step is to expand the targeted learning-rate sweep into a full per-method η and β_2 sweep for each α , including the FedAvg-style baselines. A second extension is to replace the current additive-noise oracle with a richer wireless model that includes fading, power control, and synchronization error. Finally, the MAC sweep should be repeated at stronger operating points, such as smaller α , larger γ , or smaller clipping factor k , where robust pre-processing has a clearer signal to act on.

References

- [1] McMahan B, Moore E, Ramage D, et al. Communication-efficient learning of deep networks from decentralized data[C]//Proceedings of the 20th International Conference on Artificial Intelligence and Statistics. PMLR, 2017: 1273-1282.
- [2] Wang C, Chen Z, Pappas N, et al. Adaptive federated learning over the air[A]. 2024.
- [3] Li T, Sahu A K, Zaheer M, et al. Federated optimization in heterogeneous networks [C]//Proceedings of Machine Learning and Systems (MLSys). 2020.
- [4] Karimireddy S P, Kale S, Mohri M, et al. SCAFFOLD: Stochastic controlled averaging for federated learning[C]//International Conference on Machine Learning (ICML). 2020.
- [5] Reddi S J, Charles Z, Zaheer M, et al. Adaptive federated optimization[C]//International Conference on Learning Representations (ICLR). 2021.
- [6] Kairouz P, McMahan H B, Avenet B, et al. Advances and open problems in federated learning[J]. Foundations and Trends in Machine Learning, 2021, 14(1–2): 1-210.
- [7] Wang J, Charles Z, Xu Z, et al. A field guide to federated optimization[A]. 2021.
- [8] Goldenbaum M, Boche H, Stanczak S. Robust analog function computation via wireless multiple-access channels[J]. IEEE Transactions on Communications, 2013, 61(9): 3863-3877.
- [9] Nazer B, Gastpar M. Computation over multiple-access channels[J]. IEEE Transactions on Information Theory, 2007, 53(10): 3498-3516.
- [10] Zhu G, Wang Y, Huang K. Broadband analog aggregation for low-latency federated edge learning[J]. IEEE Transactions on Wireless Communications, 2020, 19(1): 491-506.
- [11] Amiri M M, Gündüz D. Machine learning at the wireless edge: Distributed stochastic gradient descent over-the-air[J]. IEEE Transactions on Signal Processing, 2020, 68: 2155-2169.
- [12] Yang K, Jiang T, Shi Y, et al. Federated learning via over-the-air computation[J]. IEEE Transactions on Wireless Communications, 2020, 19(3): 2022-2035.
- [13] Sery T, Shlezinger N, Cohen K, et al. Over-the-air federated learning from heterogeneous data[J]. IEEE Transactions on Signal Processing, 2021, 69: 3796-3811.
- [14] Şahin A, Yang R. A survey on over-the-air computation[J]. IEEE Communications Surveys & Tutorials, 2023, 25(3): 1877-1908.
- [15] Nikias C L, Shao M. Signal processing with alpha-stable distributions and applications[M]. Wiley-Interscience, 1995.
- [16] Win M Z, Pinto P C, Shepp L A. A mathematical theory of network interference and its applications[J]. Proceedings of the IEEE, 2009, 97(2): 205-230.

- [17] Duchi J, Hazan E, Singer Y. Adaptive subgradient methods for online learning and stochastic optimization[J]. *Journal of Machine Learning Research*, 2011, 12(61): 2121-2159.
- [18] Kingma D P, Ba J. Adam: A method for stochastic optimization[C]//*International Conference on Learning Representations*. 2015.
- [19] Ward R, Wu X, Bottou L. AdaGrad stepsizes: Sharp convergence over nonconvex landscapes[C]//*International Conference on Machine Learning (ICML)*. 2019.
- [20] Défossez A, Bottou L, Bach F, et al. A simple convergence proof of adam and AdaGrad[J]. *Transactions on Machine Learning Research (TMLR)*, 2022.
- [21] Blanchard P, El Mhamdi E M, Guerraoui R, et al. Machine learning with adversaries: Byzantine tolerant gradient descent[C]//*Advances in Neural Information Processing Systems (NeurIPS)*. 2017.
- [22] Yin D, Chen Y, Kannan R, et al. Byzantine-robust distributed learning: Towards optimal statistical rates[C]//*International Conference on Machine Learning (ICML)*. 2018.
- [23] Pillutla K, Kakade S M, Harchaoui Z. Robust aggregation for federated learning [J]. *IEEE Transactions on Signal Processing*, 2022, 70: 1142-1154.
- [24] Huber P J, Ronchetti E M. Robust statistics[M]. 2nd ed. Wiley, 2009.
- [25] Hampel F R, Ronchetti E M, Rousseeuw P J, et al. Robust statistics: The approach based on influence functions[M]. Wiley, 1986.
- [26] Chambers J M, Mallows C L, Stuck B W. A method for simulating stable random variables[J]. *Journal of the American Statistical Association*, 1976, 71(354): 340-344.
- [27] Li J, Chen Z, Chong K F E, et al. Robust federated learning over the air: Combating heavy-tailed noise with median anchored clipping[A]. 2024.

Appendix

(Supplementary Materials)

A. Repository Layout and Reproduction Commands

The main experiment commands are collected in `run_all.sh` and `RUN.md`. The revised heavy-tail runs are stored under `results/heavytail/`; generated figures are under `results/figures/`; generated LaTeX tables are under `template/data/generated/`.

The core module layout is

Module	Purpose
<code>src/channel/noise.py</code>	Chambers–Mallows–Stuck $S_\alpha S$ sampler
<code>src/channel/ota.py</code>	NoisyOracle for $\tilde{g}^t = (\sum_n \Delta_n^t + \xi^t)/N$
<code>src/channel/mac.py</code>	Median Anchored Clipping pre-processing
<code>src/data/datasets.py</code>	MNIST/CIFAR-10 loading and Dirichlet partition
<code>src/optimizers/baselines.py</code>	FedAvg-OTA
<code>src/optimizers/adagrad_ota.py</code>	AdaGrad-OTA with fractional $v_t^{1/\alpha}$
<code>src/optimizers/adam_ota.py</code>	Adam-OTA with fractional $\hat{v}_t^{1/\alpha}$
<code>src/experiments/federated.py</code>	Training loop
<code>src/experiments/heavy_tail.py</code>	α ablation, MAC compare, MAC sweep
<code>src/experiments/lr_sweep.py</code>	Per-method server learning-rate sweep
<code>src/experiments/ablation.py</code>	Noise-scale and client-count ablations
<code>src/experiments/plot.py</code>	All experiment plots
<code>src/experiments/plot_mechanism.py</code>	Mechanism-diagnostic plots
<code>src/experiments/report.py</code>	LaTeX table generation

B. Consolidated Hyperparameter Table

Every experiment in the thesis uses the values below, with per-table overrides stated explicitly in Chapter 4.

Parameter	Default	Where it is overridden
Dataset / model	CIFAR-10 / ResNet-18	MNIST sanity uses MLP
Clients N	100	Client-count ablation (5/10/20/50)
Communication rounds	100	Main comparison uses 200
Local epochs E	1	Main comparison uses 5
Batch size	64	—
Local learning rate	0.01	—
Server learning rate η	0.01	Client-count ablation uses 10^{-4}
Momentum / β_1	0.9	—
Adam β_2	0.999	—
Adam-OTA ε	10^{-8}	—
AdaGrad-OTA ε	10^{-4}	—
Noise tail index α	2.0	α ablation $\in \{1.1, \dots, 1.9\}$; MAC compare 1.3
Noise scale γ	0.05	MAC sweep $\in \{0.05, 0.1, 0.2\}$
Partition	non-IID Dirichlet 0.1	MNIST sanity uses IID
Seeds	42/43/44	Single-seed cases use 42 only

C. Server-Optimizer Pseudocode

All three server methods share the per-round inputs ($\tilde{g}^t$, server learning rate η) and share the in-place update $w_{t+1} \leftarrow w_t - \Delta w$. They differ only in how Δw is computed from $\tilde{g}^t$.

Optimizer	Update rule
FedAvg-OTA	$\Delta w = \eta \tilde{g}^t$
AdaGrad-OTA	$s_t = \beta_1 s_{t-1} + (1 - \beta_1) \tilde{g}^t$; $v_t = v_{t-1} + s_t ^\alpha$; $\Delta w = \eta s_t / (v_t^{1/\alpha} + \varepsilon)$
Adam-OTA	$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \tilde{g}^t$; $v_t = \beta_2 v_{t-1} + (1 - \beta_2) \tilde{g}^t ^\alpha$; $\hat{m}_t = m_t / (1 - \beta_1^t)$; $\hat{v}_t = v_t / (1 - \beta_2^t)$; $\Delta w = \eta \hat{m}_t / (\hat{v}_t^{1/\alpha} + \varepsilon)$

Author's Biography

Peiran Wei is an undergraduate student in Electrical and Computer Engineering at the Zhejiang University–University of Illinois Urbana-Champaign Institute, graduating in 2026. During the undergraduate program, the author has worked on projects spanning federated learning systems, signal processing over the wireless multiple-access channel, and machine-learning optimization for edge intelligence.

The author's research interests focus on heavy-tail-robust optimization, analog computation over wireless channels, and adaptive federated optimization. The present thesis continues that direction by experimentally validating a scoped robustness claim of adaptive over-the-air federated learning under symmetric α -stable interference.

Project repository: https://github.com/promissa/st4001_project.